

part of a training program used to help get actors in shape for a TV series named “Spartacus.” The sessions below are timed circuits that can be done in a small space with very little equipment. They are excellent choices for crowded gyms or for traveling climbers with very little equipment. There is no magic to the sequence or the specific exercises. You should seek to do exercises you know well and can execute for the entire duration of the sets. You should back the loading off to where each set is only somewhat difficult, aiming for a slightly increasing level of fatigue as the session progresses. Again: avoid the pump and avoid getting so wiped out that you can’t train tomorrow.

For these sessions, select just one of the circuits listed below. Follow that circuit for the number of rounds and durations listed in the table at the end of the section. Stick with one circuit for an entire training cycle if possible.

CIRCUIT 1

Push-Up

Back Lunge

Plank

Renegade Row

Kettlebell Deadlift

CIRCUIT 2

2-Arm Bent-Over Row

Air Squat

Push-Press

Squat Thrust

Diver Sit-Up

T Push-Up

Kettlebell Swing

Plank

CIRCUIT 3

- Air Squat
- Side Lunge and Touch
- Push-Up
- Renegade Row
- Plank
- Back Lunge
- Kettlebell Deadlift
- Squat Thrust
- Push-Press
- Kettlebell Swing

Spartacus Progressions

SESSION #	ROUNDS	WORK TIME (SECONDS)	REST TIMES (SECONDS)	REST BETWEEN ROUNDS (MIN.)
1	2	30	30	3
2	2	40	40	3
3	3	30	30	3
4	3	40	40	3
5	3	45	30	3

SPECIFIC ENDURANCE SESSIONS

ALACTIC CAPACITY CIRCUITS

These are the same circuits I programmed in the general phase, so why put them here? Because anaerobic capacity takes a long time to develop. Yes, it would be boring for you to do the same training for 8 weeks in a row, so be sure to advance the specific phase exercises over the general phase loads. Weighted exercises should be done with more weight, ladders on smaller rungs, hangs on smaller edges, air bike at a higher wattage, etc. You can even

add more total rounds to the workouts, but do not increase work time per set or decrease rest.

Alactic Capacity circuits feature high-intensity efforts performed for 5-10 seconds, and followed by passive rest. Sport science taught us that these efforts done on a 1:5 or longer work:rest ratio were best, so for many cycles we trained by doing a 5-10 second effort each minute. What we slowly realized was that alternating the focus of the efforts from finger-specific to total-body movements allowed us to cheat the work:rest ratio a bit, and thus the intervals on a 30 second clock were born.

In the Specific Endurance phase, we begin to focus on more climbing-like efforts. Each set of intervals features 5 exercises done in sequence. You start each exercise at the beginning of a :30 repeating clock, then rest the remainder of the time. In general this gives you about 20 seconds to move to the next exercise. After the 5th exercise, you'll rest an additional 30 seconds, thus making each round 3 minutes in length. In the general phase, we did 1-2 finger-specific exercises. In this phase, we'll aim for 3 each round.

The specific template is as follows:

0:00 >10 seconds Upper Body Explosive or Edge Hang
0:30 >10 seconds Lower Body / Total Body Explosive
1:00 10 second Edge Hang
1:30 >10 seconds Upper Body Explosive
2:00 >10 seconds Upper Body Strength
2:30 rest full 30 seconds

There is not a lot of time for moving around the gym here. I suggest you grab a couple of tools and set yourself up by the Campus Board. My normal specific session is the following:

0:00 Campus Ladder 1-3-5-7-9 (on medium rungs)
0:30 6x Kettlebell Swings (~½ bodyweight bell)

1:00 10 second Edge Hang (bodyweight, 10mm)
1:30 3x Campus Doubles (large edges or jugs)
2:00 2x Power Pull-Up
2:30 rest full 30 seconds

The sessions are built on doing several rounds of 3 minutes, as described above. A series of rounds will be done back-to-back (usually 3-6 rounds per series), with a long rest between series. It is possible to change a few of the exercises between series, but don't get carried away with variety.

You must continually assess whether your strength and power are staying up. If you start to see a notable decline in output or in exercise form, it's indicative you've reached the end of the effective session length. If you've transitioned to this phase after doing these intervals in the general phase, you are going to want to back off a bit over last month's peak volume. When we move to more finger-intensive training, we tend to see an initial drop in work capacity.

Going deeper into the training will be counter-productive. In the progressions below, understand that almost all of us will hit a level where we should not add more work, and should stick with the same session for a week or two. This session is appropriate once per week. If you do decide to do this style of session more frequently, only advance the difficulty weekly, not each session.

Being a grown-up about it rather than charging forward into tiredness will produce better long-term results.

Progression is as follows:

Session 1: 2 series of 4 rounds with 5 minutes between.
(29 minutes total)
Session 2: 2 series of 5 rounds with 5 minutes between.
(35 minutes total)
Session 3: 2 series of 6 rounds with 5 minutes between.
(41 minutes total)



Session 4: 3 series of 5 rounds with 5 minutes between.
(55 minutes total)

If your form breaks down or you are unable to complete an exercise, don't follow the progression. Back off on the exercise, reduce the number of reps per set, or reduce the load. Your feeling at the end of any of these sessions should be tired and powered down, not nauseous and pumped.

As with most training, start each session with a good warm-up that gets you ready for the explosive and high-intensity work to follow. Also, the 8-10 minutes of easy total-body cardiac output work seems to really help with feeling good after a hard alactic session.

LONG T2P

T2P stands for "traverse-to-problem." This is one of my favorite specific endurance sessions in our gym, but it might not be ideal for all facilities. You'll need to have a wall that has enough hold density for low-intensity traversing, and you'll need to be able to do the traversing. If your gym is a busy one with spread-out problems this might not be ideal. The traverses need to be 25-40 moves of easy terrain or big holds, that lead you to a finish problem that is just around your onsite level.

For most sessions, you'll want to pick 3-4 problems, and then assure that you can do both leftward and rightward traverses. Traversing isn't the same as climbing up, but we do it occasionally, and we might as well balance out our leftward and rightward skills.

1. Warm-up. 5-10 minutes, bouldering volume ladder. Do an easy boulder problem, then rest 30-45 seconds. Next, do 2 easy problems with no rest between, then rest 30-45 seconds. Follow this with 3 problems back-to-back. If needed, repeat this sequence a second time. Rest 5 minutes before starting training.

2. Work Sets. Keep the intensity of the traverses low, and move through the problems as relaxed and calm as possible. Have the discipline to seek not the pump but lots of quality work, instead. Volume should progress over the training cycle, but you might not advance each time. Do not increase the difficulty of the traverse, nor reduce the rest times between efforts. The best factor to manipulate would be to add more sets, which would help increase your capacity. Secondly, you could increase the number of moves in your traverses, which may add to blood delivery capabilities and local muscular endurance. Save the pump for the power endurance sessions later in the season. Here is a suggested progression:

SESSION #	NUMBER OF SETS	REST BETWEEN SETS (IN MINUTES)	APPROXIMATE SESSION DURATION (WARM-UP + COOL-DOWN)
1	6	5	65
2	8	5	80
3	9	5	90
4	10	5	100

3. Cool-down. In most of our endurance work, more low-intensity efforts equal more capacity. Finishing this workout with 8-10 minutes of total-body general endurance (at conversational intensity) is excellent. Air bike or rowing machine.

LOW-LOAD DENSITY

Density training is a staple of muscular endurance training. Instead of trying to increase the duration of your session or to add difficulty to the work sets, you instead try to fit more work at the same difficulty into a fixed amount of time. The first step is to figure out how much work you have time to do. In these sessions, I recommend you do boulder problems, though you could conceivably do a weight circuit or series of hangs or something. This work, however, would fall more into the general endurance category...we are trying to do more climbing during this phase.

Most climbers are capable of doing 45-60 minutes of climbing in the work sets of these sessions. The set up is simple: warm-up for 10 minutes or so (I like a combination of cardiac output work and climbing), then set a timer for the planned duration, and start climbing boulders. The problems should be 2-3 grades below your onsight level. Set firm boundaries here, so you don't get sloppy toward the end of the workout and start adding in problems that are too easy just to get more mileage. Remember, quality counts.

Track the V grades of the boulders you do. At the end of your planned duration, stop the clock and add up all your numbers. Divide this number by the number of minutes in your session. This will give you a session density number. The goal of these workouts is to push that number higher. Aim to do 4-8 sessions at the most before cycling out and moving on to other training.

POWER-ENDURANCE TRAINING SESSIONS

TIMED COMBINED CIRCUITS

The Timed Combined Circuits are perfect for groups of athletes on the same general program, or for an individual looking to keep the workout duration fixed. The plan is to do two circuits of 6 exercises per circuit, with each circuit lasting 20 minutes. Most climbers will complete about the same number of rounds in any given circuit over the course of a 4-week block of training. These circuits can be advanced two ways:

1. More exercises / rounds per 20 minute period. This increases the exercise density, and thus the anaerobic endurance load of the session.
2. Heavier weights / harder movements. This increases anaerobic power more than capacity.

We set up the Timed Combined Circuits in the Power Endurance phase as follows:

CIRCUIT 1:

Upper Body Pull x6-10 reps
Lunge or Step-Up x 6-10 reps each side
Hanging Straight Leg Raise x10 reps
Explosive Upper Body Press x 6-10 reps
Edge Hang 10 seconds
Mobility 30-45 sec

CIRCUIT 2:

Campus Ladder x1 lap
Explosive Hip Hinge x10 reps

Ab Wheel or Dragon Flag x5 reps
Upper Body Pull x6-10 reps
Edge Hang 10 seconds
Mobility 30-45 sec

The pacing is key here. You want to continue to be able to do quality work, so don't go so fast that your form starts to fail or you need to reduce loads in subsequent sets. On the flip side, you do want to be working hard...if you get fewer than 3 rounds per circuit, you probably are showing a lack of capacity and should consider revisiting a general endurance phase before trying to ramp up your power endurance.

Also, you are going to have to put in a real effort to increase the work you do each session. Be sure to add load or increase density every single session for the entire training cycle. I recommend only one of these sessions per week for a 4-week cycle, while training Campus Foot-On Ladders, Density, or Rhythm Intervals the other PE training day.

RHYTHM INTERVALS

There are times in life when things fall apart and many of us drift away from doing the things we love. My love of climbing has barely wavered since the mid-80s, but at one point I couldn't make myself do it, even though I probably needed it more than ever. My good friend and next-door-neighbor, Todd Skinner saw this. He badgered me for days to walk the 50 feet to his garage and "do some movement." In the garage, we had built a 30-degree overhanging wall and it was covered with a variety of evenly-spaced, matched holds, much like a system board. When he finally got me out there, he said, "Today, we'll be in here for just 10 minutes. He got on the wall holding two small edges, moved up to another of the same size with his left, then reached back down. He did the same with the right. He continued to repeat this sequence over and over for

maybe 10 on each side. Then it was my turn. We switched back and forth maybe a dozen times and I was completely spent. “That, my friend, is maximal forearm pump with minimal body fatigue,” he said. It was enough for me to break my funk, get over to the garage for a few minutes at a time, and relight the fire.

Over the course of several years, we refined the workout into the well-known 4 minute block. 30 seconds of steady controlled up-and-down moves on small holds, followed by 30 seconds of shaking out (active rest) on a jug, repeated four times.

We use these as a maximum lactic load exercise because you don’t have to do anything except move the arms as the pump grows. You are going to want to train on a slightly overhanging wall, maybe 15-30 degrees. To perform Rhythm Intervals, you’ll simply pick a series of four matched edges (about a pad-sized edge is best) and set them up at shoulder width, 20-24” between the bottom pair and upper pair. In the middle, you’ll place a jug on which you can recover actively. This will need to be a very good hold for most climbers.

Depending on your goals or the nature of the climbing in your area, you might modify the hold selection. For Smith climbers, you might use small crimps as the “work” holds, and a large flat edge as the rest. Maple Canyon climbers might want rounded holds and a big open hand jug. The key consideration is to train the same set of grips for a whole cycle rather than trying to insert too much variety in this exercise.

These can be done easily on a System Wall, and can be completed on a Campus Board, as long as there is a big recovery jug nearby. Make the movements simple, but maintain some power in the system, meaning don’t try to stay tight to the wall and locked off on the bottom holds. You should drop to a full straight-arm position at the bottom of each rep.

Notation is as follows: 2x 4x30:30 means two rounds of 4 sets, with 30 seconds movement and 30 seconds active rest. Recovery between the rounds should be 4-6 minutes.



I suggest starting every rhythm session with about 10 minutes of Extensive Endurance climbing, going long enough that you feel physically warm and are feeling some blood in the forearms. These are somewhat short sessions, so combining them with a Timed Circuit or Complexes would be appropriate.

8 Session Rhythm Board Progression A 4-week (8 session) progression might look like this:

Day 1: 2x 4x20:40, rest 6 min between rounds.

Day 2: 2x 4x30:30, rest 5 min between rounds.

Day 3: 2x 4x30:30, rest 4 min between rounds.

Day 4: 2x 4x35:25, rest 5 min between rounds.

Day 5: 2x 4x40:20, rest 5 min between rounds.

Day 6: 2x 4x40:20, rest 4 min between rounds.

Day 7: 2x 5x30:30, rest 6 min between rounds.

Day 8: 2x 5x30:30, rest 5 min between rounds.

CAMPUS FOOT-ON LADDERS

Boring but effective. This workout is done on a Campus Board with foot rails. The idea is to do a fixed amount of simple laddering - don't worry about getting fancy here - to destroy the forearms'

local muscular endurance. The key is to recover enough between sets that you can have a usefully long session. If you pump out on the first set, your training effect will be almost nothing.

Warm-up with some bouldering and a few minutes of movement prep. I also like to do 2-4 minutes of threshold-level work on the air bike or rowing machine to really get the blood flowing and the breathing up.

Pick a set of rungs that you can ladder up and down on for at least a minute when you're fresh. This is a good place to start. If your rungs are too small...again the session duration becomes a problem.

Start by doing 20 moves on-the-minute for 10 minutes. Once you finish 20 moves, step off the board and rest until the beginning of the next minute. After your 10th set, rest about 10 minutes and stretch, ignoring your phone. Repeat the same effort, 20 moves on-the-minute for 10 more minutes. Next session, you'll advance.

To advance, you can do one of the progressions below. Do only one.

1. Increase the number of moves per set. This will aim to increase your anaerobic power. I suggest sticking with the 10:10:10 format above, but move to 22 moves per minute. Advance to 24, then 26, then 28, and so on. If you can get to where you are climbing for more than 45 seconds per minute, the difficulty of the movement should be advanced (smaller rungs or bigger reaches) and you should restart at 20 moves per minute.

2. Increase the number of sets per workout. This will increase your capacity. Do this slowly and make sure you don't get to where you "drop-off" too quickly in your ability to keep moving. If you start out at 20 moves per minute and do it for 10 sets, but in set 11 you only make 18, followed

by 16... it is too hard for you right now. Progress by one or two sets per round each workout

For example:

10 sets, rest 10 minutes, 10 sets

11 sets, rest 10 minutes, 11 sets

12 sets, rest 10 minutes, 12 sets

13 sets, rest 10 minutes, 13 sets

3. Decrease the training rung size or increase the reaches.

This change will help increase your anaerobic power. Simply making each movement a little bit harder pays big benefits in this realm. The first time you go to the next-smaller rungs, you might not be able to stick with the same 20 moves each minute. If you increase the difficulty and see yourself start to fail at, say, set 6, go ahead and stop the round, take 10 minutes' rest, and then do a second round with the goal of getting that same 6 sets. Increase from there as your fitness improves.

HIGH-LOAD DENSITY

High-Load Density is a similar session set-up to the low-load version, but the difficulty of the boulders is closer to your onsight level. The general plan is to do as many boulder problems as you can do well in a fixed amount of time, and then try to do more climbing within that same timeframe in subsequent sessions. Repeating problems is fine, but your skin, joints, and brain will thank you if you have some variety in there. This is a good session to work on incorporating skill with your power endurance, but you need to be disciplined about maintaining it - get sloppy and you'll pay for it long-term.

You'll be best-served by starting out doing a 20 minute block of

bouldering, noting the problems you complete, and then taking a 10 minute rest before doing a second 20 minute block. Stick with this format for the first cycle of density you do, and then consider building to a pair of 30 minute blocks in future cycles. You can eventually aim for a single 60 minute cycle, but watch for skill degradation toward the end of the session.

Since this is an energy-system session and not a success-or-fail session, be sure to note problems that you almost completed as well as sends. As long as you don't fall on the first move, those problems still "count" when it comes to loading your system.

Track the V grades of the boulders you do. At the end of your planned duration, stop the clock and add up all your numbers. Divide this number by the number of minutes in your session. This will give you a *session density number*. The goal of these workouts is to push that number higher. Aim to do 4-8 sessions at the most before cycling out and moving on to other training.

In high-load sessions, you'll be doing harder problems and will naturally rest more. Over a full training cycle, you might only increase your density by 2-3 problems in an hour...but remember we are looking for long-term benefits, and over several cycles you'll see a major phase-shift in your anaerobic capacity.

COMPLEXES

Complexes are short circuits of exercise performed with a single implement (or pair) that can be used as short and effective anaerobic sessions. Although these are not specific to climbing in regard to movement, you can tailor them to be very specific to intensity and duration of climbing or bouldering efforts. I include these here for climbers that have limited access to power endurance training on a climbing wall, for those who are traveling, or for those who feel they need greater total-body endurance for very physical climbing



PHOTO: JOSH LARSON

like you find in stalactite-filled limestone caves, offwidths, and modern competition boulders.

These are great sessions for a crowded gym or for a van-living road tripper with just a couple of dumbbells in the back. In each session, we try to perform at least 5 exercises and cover all the major movement patterns we can. Complexes can be done with almost any tool, but I suggest bodyweight, with dumbbells, with kettlebells, or with barbells. Below, I'll outline one example of each, though there are really hundreds of possible combinations.

Some of the exercises will feel very easy at the fixed load and some harder. Avoid the temptation to pull out several different weights to “optimally load” each exercise. The value here is in simplicity and in being able to transition quickly. In fact, if possible, you should move right from the end position of the last rep of one exercise into the start position of the next. All exercises are performed back-to-back in this way, without putting the weight down or pausing. Once you complete all of the exercises in one complex, you then take a short rest before starting the next one. You'll do several rounds of a circuit, and then you're done. Short session. Good results.

Although you all know by now there are many ways to progress training, you want to be careful to stay within the bounds of training for climbing-like loads and durations. For example, you could conceivably add many exercises and reps to one circuit, but how long do you really need to be under load in a typical climbing situation? Even very long routes give us some kind of recovery, so getting to the point that you can go hard for 40 minutes non-stop is probably beyond the point of useful. Likewise, reducing rest can be an attractive way to “make it hard,” but we risk having to reduce the workload to a point that it no longer satisfies the desired level of intensity.

Progression

SESSION #	EXERCISES	ROUNDS	REPS PER EXERCISES	REST BETWEEN ROUNDS
1	5	3	5	:60
2	5	4	5	:50
3	5	5	5	:45
4	5	5	6	:45
5	5	5	8	:45
6	5	5	10	:45

BODYWEIGHT COMPLEX

Air Squat

Push-Up

Glute Bridge / Hip Lift

Pull-Up or Inverted Row (if no implements are available, get creative)

Hardstyle Plank (hold for 15 seconds)

SINGLE KETTLEBELL COMPLEX

Swing

Goblet Squat

Press Left Hand

Press Right Hand

Row Left Hand

Row Right Hand

Weighted Sit-Up

TWO KETTLEBELL COMPLEX

2KB Swing
2KB Clean
Rack Squat
2KB Press
Renegade Row (alternating)

TWO DUMBBELL COMPLEX

Deadlift
Power Clean
Front Squat
Push Press
Renegade Row (alternating)

BARBELL COMPLEX

Romanian Deadlift
Row
Front Squat
Push Press
Roll-Out (use barbell as ab wheel)

Make substitutions as dictated by your experience and skill level. Once you have a good complex that flows well, start through the suggested progressions above. Typically, you'll only do one complex workout per week in the power endurance phase, so you may only make it to session 4 on the table. This is a good session on its own or can be combined with Foot-On Ladders or Rhythm Intervals.

PERFORMANCE PHASE TRAINING SESSIONS

When you are performing, you want to go out and climb the best you can. You built everything up for this part of the year, and you need to respect the work you put in. You could write a whole book on mindset, tactics, motivation, and techniques for sending during this phase, but that is beyond the scope of what we are covering. For this part of your training, I recommend you read *The Rock Climber's Training Manual* by Mark and Mike Anderson. Their section on tactics is the best I've ever seen.

For the sessions in this phase, you need to “keep the goal the goal.” Each session within the phase can simply be a reduced-volume version of the strength, power, or endurance work you did earlier in the year. This serves two purposes: First, you are familiar with the flow of the sessions and can easily figure out warm-up details and equipment needs. Second, you will have good records of what you did in training earlier in the cycle, so you will be able to tell if your numbers are still good.

All things being equal, you should look to build your in-season efforts around maintaining power. This is the most relevant system to doing hard movement, and is hard to keep up without regular practice.

I suggest dropping the session volume by around 50% to start with. Some of us tend to freak out if the gym sessions don't seem hard enough, but again I'll remind you that you want them to be easy as possible so you can go hard at the crag. Here is an example Integrated Boulder session done during a Specific Power phase:

1. 5-10 minutes of a standard Bouldering Warm-Up

2. 4 rounds of:

Power-Oriented Boulders V6, rest 1:30, V5, rest 1:30,
V6, rest 1:30
5x Ball Slam@16 pounds, rest 30 sec
5x Box Jump 30", rest 30 sec
3x Power-Pull-Up @ BW +8 pounds, rest 4 minutes.

3. 4 rounds of:

Power-Oriented Boulders V5, rest 1:30, V6, rest 1:30
5x Power Push-Up, rest 30 sec
5+5x Kettlebell Snatch @53 pounds, rest 30 sec
2x Power-Pull-Up@ BW, rest 4 minutes.

In the Performance Phase, you could maintain basically the same intensity, but drop the total volume of the session:

1. 5-10 minutes of a standard Bouldering Warm-Up

2. 2 rounds of:

Power-Oriented Boulders V6, rest 1:30, V5, rest 1:30,
V6, rest 1:30
4x Ball Slam@16 pounds, rest 30 sec
4x Box Jump 30" , rest 30 sec
2x Power-Pull-Up @ BW +8 pounds, rest 4 minutes.

3. 2 rounds of:

Power-Oriented Boulders V5, rest 1:30, V6, rest 1:30
4x Power Push-Up, rest 30 sec
4+4x Kettlebell Snatch @53 pounds, rest 30 sec
2x Power-Pull-Up@ BW, rest 4 minutes.

The workout is still hard even with the reduced volume, but you will be leaving the gym a lot fresher than when you did in the Power phase. Once performance begins to decline at the crag or when you wrap up your month-long road trip, you can and should take a week or two to recover and reset, then get back to work on your next training cycle.

MAINTENANCE SESSIONS DURING TRAINING PHASES

There is a lot of worry among hard-training climbers that their fitness gains will be lost if they stop or reduce training any facet of their fitness. Remembering that going out and doing the sport is very good for your conditioning and your strength - yes, there are highly successful climbers who don't train at all - you don't have to worry about too much drop-off. In this plan, you are going to address training non-focus fitness once every three sessions - which is not enough to develop greater fitness, but is certainly enough to maintain it.

To review: During strength-focused phases, you need to build a maintenance session that helps you hold on to your power as well as some of your endurance. In power-oriented phases you need to maintain endurance and strength, and during endurance phases, you need to do a short, focused session of strength and power. It can be easy to get hung up on specifics and durations, but the main keys are as follows:

1. Sessions should be no more than 90 minutes.
2. Loading should be intense - in the range of 75%-90% of your numbers during the phase where you focused on strength, power, etc. (If your bench press max reached 180 in the strength phase, maintenance loading should be in the 135-160 range)
3. Volume should be about $\frac{1}{3}$ to $\frac{1}{2}$ what you did in-phase. If you were doing 4 sets of pull-ups, you'd drop to 1-2 sets per workout. If you built up to 60 minutes of EE interval combos, you'd want to drop to 20-30 minutes.

4. Remember that it won't feel like enough, based on the volume you did during phases focused on the systems you are now only maintaining. It's enough.

POWER AND ENDURANCE MAINTENANCE

During general and specific strength phases, you should aim to maintain power and endurance during one short session a week. Both of these values are relatively easy to hang onto - you'll be doing a few powerful movements in many of your strength sessions, and you'll also be climbing or bouldering, which both help to hold power levels high. Endurance maintenance is much the same. Keeping your total workload up will help hold your capacity high. Going out and climbing on the weekends will help, too. Thus, a weekly session is enough.

Sessions can start with a short 10-15 minute opener of several easy boulder problems, increasing in difficulty. I like to do an ascending ladder, such as:

V0, V1, rest

V0, V1, V2, rest

V1, V2, V3, V4, rest

V1, V2, V3, V4, V5

This is followed by a similar sequence on the hangboard. Using a 20mm or larger edge:

Hang 1 sec, then drop the hands to the sides for a moment. Repeat with a 2 second hang, then a 3, and continue up to a 10 second hang...all at bodyweight, of course.

You then move into a very short power-focused circuit, such as in the Power 3x2 session in the General Power phase, but at reduced volume:

2 ROUNDS:

Box Jump x5-8 reps

Power Push-Up x5-8 reps

Hip Mobility (Frog or Firelog) x60-90 seconds

I follow this with 2-3 laps on the campus board before moving onto endurance. Keep maintenance in mind and don't try to set any records.

For endurance, a simple short combination of EE Interval Combos is great. Climb on easy terrain for 4-6 minutes, transition to a total-body cardiac output exercise (SkiErg, Rower, or Air Bike) and do that for a similar duration, and alternate back and forth for 2-3 rounds.

Don't feel like you need to take this session as gospel, but rather as an example. Low volume is the big key. Don't go hard. Maintain. Then put your tough face on and get ready for the strength training to come.

STRENGTH AND POWER MAINTENANCE

During endurance phases, you have got to maintain strength and power. This is the biggest crime of classical periodization: letting go of high-threshold athleticism during conditioning phases. Because of the high workload and fatiguing nature of endurance, it is really easy to let strength and power go - looking at doing a high-intensity session when you're really tired is pretty hard. Although there is a bit of strength and power involved in some of the endurance work, you just can't get away from at least one maintenance session per week where you avoid fatigue and focus on high outputs.

The preferred session format is Integrated Bouldering, as described below. This is modeled after "Integrated Boulder A," from the Power phase, but we adjust the volume down a bit and change a few movements. Remember, too, that your intensity should drop slightly from your best numbers in-phase. We also pull in a bit of pure strength rather than just explosive power. It is minimal, but necessary.

1. 5-10 minutes of a standard Bouldering Warm-Up

2. 2 rounds of:

3x Power-Oriented Boulders, Onsight Level or so,
resting as needed between

5x Ball Slam, rest 30 sec

5x Box Jump, rest 30 sec

3x Power-Pull-Up, rest 4 minutes.

3. 2 rounds of:

2x Campus Ladder, small rungs, single-spacing

5x Power Push-Up, rest 30 sec

2x Front Squat, rest 30 sec

2x Heavy Pull-Up, rest 4 minutes.

The specifics don't matter as much as the idea (I know I keep saying this, but I get many emails per week about minutiae of training numbers, likely from people who spend more time planning training than training). A few short sets of power, a few short sets of strength. Recover, and get back to endurance.

Many climbers worry about loss of finger strength from the backoff on hangboarding. I suggest you test your numbers at the beginning of an endurance phase, test again at the end of week 2, and again at week 4. If you are still within 5%, don't worry about it. I have never seen an athlete that lost finger strength more than 5% in an endurance cycle that included the session structure above at least once every 10 days.

ENDURANCE AND STRENGTH MAINTENANCE

When we are focusing on power, we recover a lot between exercises and sessions. Because of this, we really need to have a quality maintenance session that keeps our capacity up. This session should include a bit of pure strength training, but more critically will include a lot of endurance. You might know the research that

show that training strength and endurance in the same session results in compromised strength gains. It's true, but please keep in mind that the goal here is to maintain a couple of facets of your climbing fitness and not a gain in strength. In some fantasyland, climbing training might be less complex and the demands of the sport less challenging...but then we might not love it so much.

Warm-Up: Cardiac Output 10-15 minutes + Bouldering Ladder 5-10 minutes

1. Max Hangs + Max Pulls (3 rounds:)

Pull-up x 5 (@75%)

Hang, medium edge x 10 seconds (@75%+)

DB Row x 5+5 (@75%)

Hang, medium edge x 10 seconds (@75%+)

- Rest 1-3 minutes between each exercise.

2. Long T2P

4-6 traverses into onsight-level problems, resting 5 minutes between. Adjust rest to make sure each traverse is started somewhat fresh.

3. Extensive Endurance or Volume. Do 10-20 minutes of conversational-intensity traversing or climb a couple of continuous, easy to medium level routes. Don't get pumped.

You should leave the session feeling like you could have done more, almost to the point that it seemed like a waste. Perfect. Remember that this is not your focus, and the easier this feels the harder you can go in the next power session.

PRACTICE

“You can practice shooting eight hours a day, but if your technique is wrong, then all you become is very good at shooting the wrong way.”

- Michael Jordan

The term “training” is over-used. People refer to any and everything they do in climbing as training, yet training is a very specific act. It is repeated efforts at improving specific parameters of your fitness. Imagine “training” a dog but just randomly giving it orders with no expectation of behaviors. Organized, progressive, repeated... these terms are how training must be defined.

Yes, there is training for climbing. It is done in the weightroom, on the hangboard, against the clock. And although bouldering, climbing routes, and even system board work can be considered *training*, they are really *practice*. Training is how you get stronger. Practice is how you get better.

The saying “practice makes perfect” is known to all of us, but the truth is that *practice makes permanent*. Simply doing something a lot doesn’t make you good at that thing. What we really need to do is practice correctly, do it a lot, and then reap the rewards. This is where my 75/25 rule comes back into play: 75% of your “training” time should really be practice time. Don’t go to the gym to get tired, go to get better.

GET BETTER AT GETTING BETTER

“

PEOPLE OFTEN ARRIVE AT AN ‘OK PLATEAU,’ A POINT AT WHICH THEY STOP IMPROVING AT SOMETHING DESPITE THE FACT THAT THEY CONTINUE TO DO IT REGULARLY. THE

SECRET TO IMPROVING AT A SKILL IS TO RETAIN SOME LEVEL OF CONSCIOUS CONTROL OVER IT WHILE PRACTICING TO FORCE ONESELF TO STAY OUT OF AUTOPILOT.”

DOUG LEMOV, *PRACTICE PERFECT*

When you first start climbing everything is practice. You learn new skills and motor patterns so intensely that you improve with each and every session. Over time, you become adept at the frequently-used skills of climbing - they become automatic - and your progress slows. Occasionally, you’ll learn a more subtle skill, get a bit better conditioned, or make other small breakthroughs... but eventually, most of us end up with a fixed skill set and simply alternate between fitness and non-fitness.

The more experienced we are, the longer we’ve been doing the same things, the more critical practice, and practicing right, becomes. I won’t lie: restarting the practice engine that you developed as a novice is hard to do. It’s a critical step, though. As Jonathan Harnum writes in *The Practice of Practice*, “There is no such thing as maintenance. If you’re not getting better, you’re getting worse.”

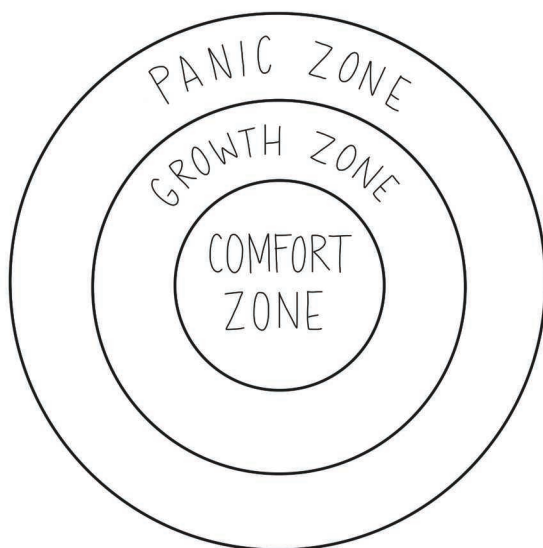
There are three basic steps to building successful practice that transcend all disciplines. Once we master these steps, getting better gets simple.

1. Practice Success

How many times have you seen it - someone at the gym or crag absolutely flailing and somehow convinced they are getting better? Sure, failure is an important part of progress, but too much failure will slow you down. A great way to look at this is the graphic of the *Zones of Comfort*. As explained by Noel Tichy: “We can never make progress in the comfort zone because those are the activities we can already do easily, while the panic zone activities are so hard we don’t even know how to approach them.”



PHOTO: NATE LILES



The sweet spot is the growth zone, where you might fail occasionally, but most of your attempts are successful. (As a side note, this is what we call “second tier” in redpoint climbing and bouldering; grades that push you to try very hard, but are achievable in one to three tries.) The vast majority of what you do in your practice and in training should allow you to complete the exercise successfully.

2. Stay Fresh

Skills degrade with fatigue. Yes, you can train to a pump, to exhaustion, or to puking, but you’re not going to acquire strength or skill by doing so. If you’re going to practice effectively, you have to be fresh for every effort. With this in mind, you have to choose wisely when it comes time to practice. A strength session with plenty of rests between efforts might be a great time to practice, where an endurance session in which you are accumulating massive fatigue will not be.

Focus your session around building skills first, then around strength, then energy system development tasks such as

endurance. If your session doesn't call for one of these facets, it's still a good idea to keep the same order.

3. Keep your Head in the Game

In order to get better, you have to look for errors...not just climb. This is what we call *active learning*. The better you get, the more you'll have to pay attention to what needs fixing. In fact, most of your practice should be about exposing errors and weaknesses so that you can correct them. This doesn't mean just working weaknesses, either. You'll also want to hone in on fine tuning your strengths.

STRUCTURE OF A PRACTICE SESSION

When we go to the gym, most of us follow a set pattern. A typical one is to shoe up, do some easy pitches or problems, slowly increase the intensity until we feel like we're working, then stay at the same trajectory until fatigue stops the session. Imagine learning math this way, or practicing music, or kicking a soccer ball. Framed in a logical sense, many of our climbing training practices seem (and are) ridiculous.

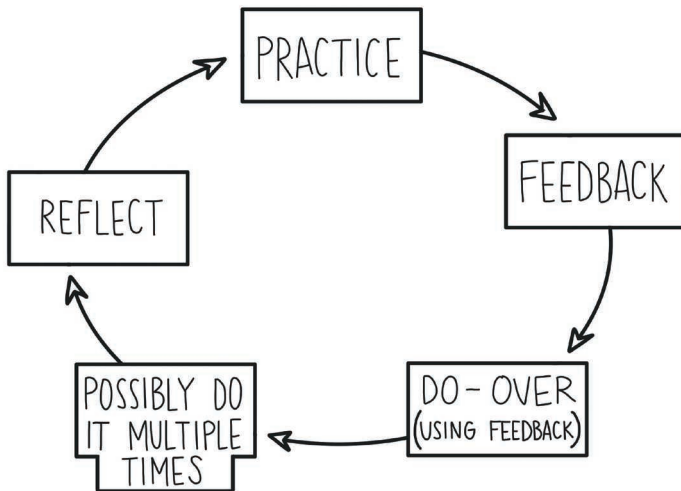
You have got to look at movement as a skill, then learn the skill. This is how effective technical practice takes place in sports.

In the excellent book *Practice Perfect*, the authors suggest that a normal practice session often takes this sequence:

1. Practice
2. Feedback
3. Reflect and discuss
4. Possibly do over

They suggest a more effective practice session would be structured like this:

1. Practice
2. Feedback
3. Do over (using the feedback)
4. Possibly do it multiple times
5. Reflect



The “Do over” step is absolutely critical. Research shows that if you do a skill right once and wrong once it is engrained in your neural circuitry equally. This means that you may or may not have acquired the skill. Repeating the correct skill multiple times is the only way of engraining the correct pattern. Understanding this one concept could be a career-changing move for a rock climber if he’s in the habit of working sections of climbs too quickly. Once you figure it out, do it again before you move on.

FEEDBACK



The feedback part of the above list can't be ignored. This is where egos get roughed-up. Take it from me...it's hard for a 35-year veteran of the sport to take criticism well. The truth can hurt, but making breakthroughs is worth the pain. For effective feedback to take place, you'll need a coach or qualified training partner to look specifically at the skill you are practicing and provide feedback as it relates to that skill. The feedback should be brief, clear, and actionable within the session. "Try to keep your hips over your feet" is usable feedback, "You need better shoes" is not.

A coach or training partner might be ideal, but you can't always have your coach on hand. You do, however, always have your phone. Videoing yourself exercising or climbing is embarrassing and may be seen as weird or egotistical, but it's educational. One step better than straight video is an app such as Coach's Eye where you can compare performances side by side. Either way, watching yourself perform can be some of the fastest learning and correction you can do.

SEPARATE PRACTICE FROM PERFORMANCE

“

ONE OF THE CHALLENGES I FACED DURING PRACTICE WAS THE DISTRACTION CAUSED BY A PLAYER'S NATURAL INSTINCT AND DESIRE TO SCORE BASKETS OR GRAB REBOUNDS. EITHER URGE IS SUCH A POWERFUL SIREN SONG THAT IT'S HARD TO MAKE THEM PAY ATTENTION AND LEARN THE 'DULL' FUNDAMENTALS THAT ENSURE SUCCESS IN SCORING AND REBOUNDING - SUCH THINGS AS PIVOTING, HAND AND ARM MOVEMENT, AND ROUTES OF PLAY.”

JOHN WOODEN

You can't learn and perform at the same time. This can be a problem. When we climb, we usually want to be out sending hard stuff, but bringing your mind back to practice will get you good faster than just going climbing. This is the beauty of both working hard routes and climbing easier things - the former forces you to learn and practice in earnest, while the former gives you an opportunity to solidify your skills. The place you tend not to learn is during redpoints - when you're sending it's *game on* and there's not a lot of time for reflection and feedback.

Since redpointing is the name of the game, you need to simply separate your foci. When you're going for it on a climb, go. When you're not sending, bring your mind into the process of learning. Even on the easiest terrain, there is always something to improve.

KEEP IT SHORT AT FIRST

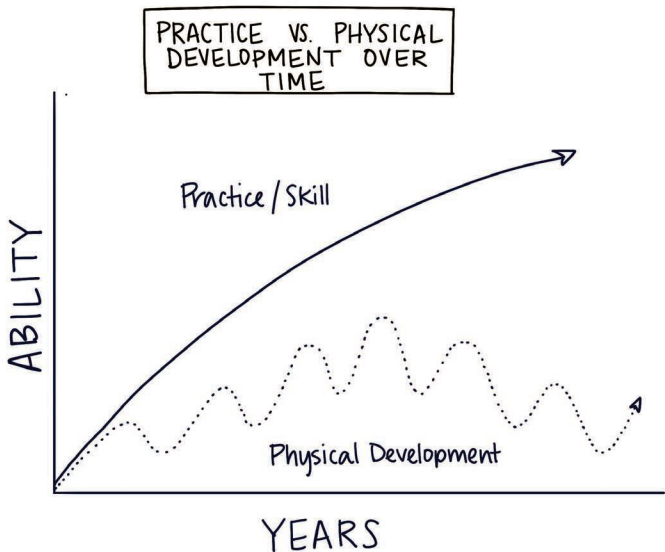
Getting back into practicing is tough. When bringing a focus on practicing movement back into your training sessions, start with just a few minutes of focus on one simple skill. As the weeks

progress, you'll find yourself able to practice longer and with more success. Eventually, movement practice will find its way into most of your sessions, and, once again, you'll start creeping up through higher and higher grades...just like in the old days.

It's important to look at climbing as an amalgamation of discrete skills that, when combined with strength and conditioning, lead to good climbing. If our overall goal is to climb well, we can break "well" down into specific pieces - skills - that we can develop to improve our overall ability. There are many ideas on how complex or "big" a skill should be in order to effectively develop it. The simple truth is that you should continue to break it down into smaller and smaller parts until you can see noticeable improvements over a fixed period of time.

A good place to start is to dedicate two weeks to developing one or two skills. These skills need to be addressed every climbing day. Give yourself two weeks, and if the skill hasn't improved, simplify it.

Remember that a beginner has the most to gain from practice. Luckily, the practice comes naturally, and there is no "autopilot"



to fall back on. Everything he does is practice. More experienced climbers, let's call them "Level 2," will see more progress from improving physical capacity than skills once they reach a passable level. Yet once a climber reaches close to his genetic potential for strength or endurance, skills once again become the most important factor in improvement.

STRUCTURING A PRACTICE SESSION



When you're practicing a skill, it's critical to have your mind in the game. Therefore, we practice skills early in the session, when our minds and bodies are freshest. I like to keep the "Five S Formula" in mind when putting together a practice:

- Simple - The skill needs to be something you understand and can easily implement in your sessions. You shouldn't need specialized tools or set-ups...otherwise you risk not doing the practice.
- Segmental - You should have a large-scale framework for improving an ability that you can progress through as you master smaller skills along the way.

- Sequential - Introduce more advanced skills at the right time and in the right order.
- Strategic - Address the biggest things that are preventing your overall progress. A training partner or coach will be helpful in deciding what this is.
- Supported - You will be more successful in practice if you have a training partner or coach to help you work through the process.

Development of discrete skills in climbing has been explored from time to time in articles and in training books. The important thing is to realize that skill in climbing is complex and that learning complex skills requires simplifying the skill into learnable chunks. Let's use the example of turning the hips on steep rock. When you first address overhanging rock as a novice, this type of movement seems strange and counterintuitive. Over time, we realize that this is a fundamental skill used in the majority of moves you'll make on overhangs. To address this skill, you might go through a process like this:

- Practice turning the hips on vertical terrain or on a system wall.
- Set up a pair of footholds to practice backstepping with a reach move for the hands.
- Using good holds on a 45 degree wall, alternate between driving left side and right side of the hips toward the wall.
- Climb out an overhanging wall on big holds, trying to drop knee on each move.
- Do the same drill as above, but make the accentuated effort of driving the hips to the wall on each move.

Daniel Coyle wrote the fantastic book *The Talent Code*, which explores talent hotbeds worldwide and attempts to break down why, exactly, certain schools, or cities, or coaches, turn out an inordinate number of talented individuals. He then followed this book with *The Little Book of Talent*, which highlights actual skill-building tactics he observed in his research for the *Talent Code*. There are a number of gems in this book, and a few of the insights he had are extra-useful for developing climbing skill. Here are some:

STARE AT WHO YOU WANT TO BECOME

I've talked about the value of spending loads of time watching climbing videos before. Yes, they are entertaining, but watching (carefully and attentively) high-level climbers can be a way of imagining your future self, which is a critical mental aspect to high performance. Spending 10-20 minutes a day watching videos of hard climbing, before practice or before bed, can have a profound effect on the quality of your practice.

BE WILLING TO BE STUPID

Coyle wrote about how Wayne Gretzky would fall down when doing solo skating drills for hockey. Was it because he sucked? No, it was because he was pushing his limits continually. It's easy to let your ego come in when you're climbing. Years ago, I remember climbing around a top-level climber who flat refused to get on any 5.12s - even though it was clearly the grade range she should have been climbing. She would warm-up and then spend the whole day working this or that 5.13...because it was *OK to fail* on the higher grade. The problem was that she would have probably seen better improvement on 5.12, which would have been her *learning zone*. Which, in contrast to what did happen, might have led to her sending.

The opposite strategy is equally problematic - always climbing on routes you are comfortable with, or are in a range where you're sure you won't fail. The great learner is going to spend a lot of time flailing on rock types, moves, and holds where he's weak - "looking stupid" - until he masters those skills.

DON'T FALL FOR THE PRODIGY MYTH

My friend BJ Tilden is a better climber than I am. He has a ton of natural talent...or does he? I've climbed with him for over 25 years, and for the first one or two, I was the better climber. Looking at it objectively, I can see that he climbs more than I do, has consistently tried harder things, and has believed in his ability to do those things. Most importantly, when the chips are down, he gives it all where I usually say "take." Which of those are God-given talents? None. Which ones can be learned and trained? All of them. Understanding that the best climber might just try harder than we do is a critical step in letting ourselves get better, too.

EACH DAY, TRY TO BUILD ONE PERFECT CHUNK

Coyle talks about building an "SAP" with each practice - a smallest achievable perfection. To do this, you've got to break your skill down to small chunks, and work on one thing only until you see a noticeable improvement. A drill such as a foot placement accuracy drill or slot deadpointing are good examples: you know what it takes to do the skill correctly, and you can detect small changes in your performance. It might seem silly to focus on such tiny changes, but that's the way all of our skills are built, anyway. All you're doing by focusing is deciding which skills your body picks up most quickly.

SLOW IT DOWN (EVEN SLOWER THAN YOU THINK)

Speed, in learning new skills, leads to sloppiness. You've seen the teenage boys during their first days at the rock gym - all power, no technique. Super-slow movement lets us avoid errors by giving us time to ingrain good patterns and feel bad ones. Working on moving slowly through steep terrain assures excellent positioning, where relying on dynamic movement might prevent it. Slow footwork teaches precision, fast footwork should be reserved for speed climbing.

Try the Kontrast speed-slow drills outlined in *Gimme Kraft*: do a problem at normal speed, then do it as fast as you can, then do it as slow as possible. What did you learn?

In many ways, your mindset in developing skills needs to be the opposite of what it is when training strength. You need calm, not psych. You need to absorb rather than produce. You need to look for microscopic improvements instead of jumps in performance. I am not arguing that you need to abandon training, but rather that you need to accept that physical conditioning will only get you so far. To excel completely in climbing, regular, deliberate, and utterly boring practice is fundamental.

PRACTICE REFERENCES

Talent is Overrated <http://amzn.to/1ef6giC>

The Little Book of Talent <http://amzn.to/1RMPNyY>

Practice Perfect <http://amzn.to/1KpCPYZ>

The Practice of Practice <http://amzn.to/1KpCW6O>
